

The CHLA **DILUTE** Insulin Calculator will accurately determine the correct dose of **dilute (10 units/mL)** rapid-acting insulin to be administered based on a specific patient order, the grams of carbohydrates eaten, and the measured blood glucose. Dilute Insulin is generally used in very small patients who are very sensitive to insulin. Because Dilute insulin is only 1/10th the concentration of standard U-100 insulin, doses are expressed in Syringe Units based on markings on an insulin syringe.

Insulin/Carbohydrate Ratio:

- 1 Syringe Unit of insulin covers **X** grams of carbohydrates eaten
- range: 1.5 – 40 gm: 1 Syringe Unit
 - ✓ ordered value; click to confirm → ✓
 - ✗ indicates an ordered value that should be clarified

Correction Factor:

- 1 Syringe Unit of insulin reduces a Blood Glucose elevated over Target Blood Glucose by **X** mg/dL
- range: 10 – 200 mg/dL: 1 Syringe Unit
 - ✓ ordered value; click to confirm → ✓
 - ✗ ordered value should be clarified

Target Blood Glucose:

- blood glucose target above which correction must be applied
- range: 130 – 300 mg/dL
 - ✓ ordered value; click to confirm → ✓
 - ⚠ ordered value is outside the usual range; clarify?
 - ✗ ordered value should be clarified

**The calculator is not enabled for input of carbohydrates eaten or measured blood glucose until all above parameters are confirmed by clicking the grey check*

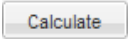
Carbohydrates Eaten:

- Enter the grams of carbohydrates eaten
- calculator will accept any value from 0 – 99 gm
 - ✗ unacceptable entry which must be corrected

Blood Glucose for Calculation:

- *Enter the current glucose measurement*
 - calculator will accept any value from 1 – 2,000 mg/dL
 - ⚠️ BG > 600 mg/dL; verify and correct if data entry was incorrect
 - 💧 BG < 70 mg/dL: hypoglycemia
 - STOP – Calculator will not calculate dose
 - DO NOT GIVE INSULIN
 - requires rescue intervention for low BG
- ❌ unacceptable entry which must be corrected

To complete calculation of dose to administer:

- *Assure all parameters and patient values are valid*
- Click  button
- Dose will be displayed, ***rounded*** to closest Syringe Units, here:

Calculated Dose to Administer



- Transfer calculated dose to MAR after 2RN check and dose has been administered
- **REMEMBER**
 - *Change time from default to time actually administered*
 - *Change dose from “1 dose” to “X Syringe Units” actually administered*